

# Prompt: Build a Unified Road & Footpath Condition Platform

Use this prompt as-is (or lightly edited) to brief Claude on building the platform. It's written to be dropped into a fresh conversation.

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## PROMPT START

You are building **RoadWatch**, a webapp that unifies three independent data sources into a single map-based view of road and footpath condition, and produces a defensible, standards-referenced condition rating per road segment.

### 1. Core objective

For any given road or footpath segment, the platform must show:

- **Current status** — a condition score/grade and the evidence behind it
- **Open issues** — potholes, cracking, erosion, missing footpath, waterlogging, encroachment, etc., each geotagged

- **History** — a timeline of reports, repairs, and condition change over time

All three data sources below must be fused into one segment-level record, not shown as three separate layers.

## **2. Data sources to integrate**

### **A. Web-scraped / public data**

- Municipal corporation grievance portals, PWD tender/work-order data, RTI disclosures, news reports of road complaints
- Treat as **low-frequency, high-authority** signal (confirms government awareness/action, gives repair history)
- Must respect robots.txt and ToS per source; flag any source that requires special handling instead of silently scraping it

### **B. User-submitted reports (including anonymous)**

- Photo + geolocation + category + optional severity vote
- Anonymous submissions must be rate-limited and abuse-resistant (device fingerprint or similar, not identity) to prevent spam/gaming without breaking anonymity
- Duplicate/near-duplicate detection by geo-proximity + time window + category before

creating a new issue record

## **B2. User rating (distinct from issue reporting)**

- A user can rate a segment directly (e.g., 1–5 stars or a slider) *without* filing a specific issue – this is a subjective "how does this road feel to use" signal, separate from the objective distress-based score in Section 4
- One rating per user/device per segment per time window (e.g., 30 days) to prevent brigading; anonymous ratings allowed but weighted lower than the objective distress score in the composite
- Store rating distribution (not just mean) so a segment with 50 ratings clustered at 2★ reads differently from one with 5 ratings averaging 2★
- Treat this as a **perception layer** that sits alongside – not inside – the ISO-referenced condition score; the UI should show both ("Measured condition: 42/100 (Poor)" and "Commuter rating: 2.3★ from 118 ratings") rather than collapsing them into one number, since a technically poor road that's still passable will diverge from a technically fine road with bad lighting/safety perception

## **C. Free satellite/aerial imagery**

- Sentinel-2 (10m, free, ESA Copernicus) for macro change detection (new construction, large-scale

surface change, flooding) – NOT resolution-sufficient for potholes

- Where higher resolution is needed, note that free options are limited (occasional Planet/Maxar open datasets, government GIS portals) – be explicit about this resolution ceiling rather than overselling what satellite data can detect
- Primary satellite use case: corroborating "when did this stretch last get resurfaced" and detecting broad-area degradation trends, not per-pothole detection

### 3. Fusion logic

Design a **segment model** (a road broken into short GPS-anchored stretches, e.g., 100–200m) as the unit that all three sources attach to. For each segment, maintain:

- `current_score` (objective, ISO-referenced – see Section 4)
- `commuter_rating` (subjective – mean + distribution + count, from Section 2B2)
- `open_issues[]` (deduplicated, sourced, timestamped)
- `history[]` (score changes, repairs logged, rating trend, source of each change)
- `confidence` – how many independent sources agree; a segment with only one anonymous report

should be visually distinguished from one confirmed by scraped municipal data + multiple user reports

#### **4. Condition rating methodology (ISO-referenced)**

Do not invent a rating scale from scratch — ground it in existing, named engineering standards, and cite which one is doing what:

- **Structural/surface condition:** base the scoring rubric on the *Pavement Condition Index (PCI)* methodology (ASTM D6433 — the internationally recognized cross-reference standard; ISO doesn't own PCI outright but this is the standard practitioners treat as authoritative) — distress types (cracking, potholes, rutting, raveling), severity (low/med/high), and density roll up into a 0–100 PCI-style score.
- **Ride quality:** reference the *International Roughness Index (IRI)*, standardized under ISO 8608 and World Bank guidelines, where sensor/accelerometer or smartphone-based roughness estimates are available (e.g., from crowdsourced phone accelerometer data during user reports).
- **Surface texture/skid resistance** (if in scope): ISO 13473 covers texture measurement methods —

note as a stretch goal, not MVP, since it needs specialized measurement.

- **Whole-body vibration exposure** (optional, footpath/vehicle comfort angle): ISO 2631 as a secondary comfort-oriented metric.
- Combine into one published composite score (e.g., 0–100 or A–F) with a transparent formula and a visible breakdown ("why this score") — do not present a black-box number.
- Explicitly separate **data confidence** from **condition score** — a bad road with thin evidence should not look the same as a bad road with strong multi-source confirmation.
- Explicitly separate **commuter rating** (Section 2B2) from **condition score** — decide whether the rating ever nudges the composite (e.g., as a tiebreaker or a "flag for re-inspection" trigger when they diverge sharply) or stays purely informational, and state that choice rather than silently blending a subjective star rating into an ISO-referenced measurement.

Ask Claude to propose the exact weighting formula and defend the choice, rather than assuming one.

## 5. Technical scope for this build

- Stack: React + Vite + Tailwind + Leaflet.js for the map, Supabase for backend/auth/storage,

deployable on Vercel/Railway, target under \$50/month infra for MVP

- Anonymous reporting flow must work without forcing signup
- AI image tagging (categorize issue type/severity from photo) – Claude Haiku via API is the intended model for this
- Geolocation-based authority tagging (ward/MLA/MP) needs real boundary data – DataMeet / india-geodata shapefiles + Turf.js point-in-polygon, not hardcoded
- Government/media-facing transparency dashboard as a separate view from the citizen reporting view

## **6. What I want from you**

1. A segment/data model (schema) that supports the fusion described above
2. The scoring formula with worked example on a sample segment
3. An architecture diagram or description showing how scraper output, user reports, and satellite data land in the same segment record
4. Flag any part of this (especially satellite resolution limits, scraping legality per source, or anonymous-abuse prevention) where the free/low-cost

constraint meaningfully limits what's achievable,  
rather than glossing over it

**PROMPT END**